

Chapter 8: Area and volume

13 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Measurement and space: MA4-LEN-C-01 (length and circumference), MA4-ARE-C-01 (area) and MA4-VOL-C-01 (volume and capacity).

CHAPTER

8

8.01 Units of measurement

8.02 Perimeter

8.03 Circumference of a circle

8.04 Area units

8.05 Area of squares and rectangles

8.06 Area of a triangle

8.07 Area of a parallelogram

8.08 Area of composite shapes

8.09 Prisms, cross-sections and views

8.10 Volume units

8.11 Volume of a rectangular prism

8.12 Volume of a prism

8.13 Capacity

How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO

Grey boxes. Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO

Blue boxes. Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL

Skill drill pages. After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
8.01	Units of measurement	3		☐
8.02	Perimeter	7		☐
8.03	Circumference of a circle	13		☐
8.04	Area units	19		☐
8.05	Area of squares and rectangles	23		☐
8.06	Area of a triangle	29		☐
8.07	Area of a parallelogram	35		☐
8.08	Area of composite shapes	41		☐
8.09	Prisms, cross-sections and views	47		☐
8.10	Volume units	53		☐
8.11	Volume of a rectangular prism	57		☐
8.12	Volume of a prism	63		☐
8.13	Capacity	69		☐

In this chapter you will

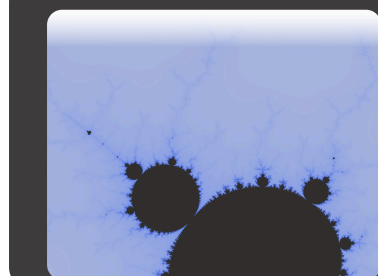
- convert units of length, mass, capacity and time
- find perimeter and circumference
- convert area units and find the area of rectangles, triangles, parallelograms and composite shapes
- draw cross-sections and views of prisms
- convert volume units and find the volume of prisms
- find capacity in litres

About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the thin branches (dendrites) that grow from the very top of the set. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.



LESSON
8.01

Units of measurement

Date _____

Learning intentions

We are learning to:

- convert between units of length, mass, capacity and time

Success criteria

I can:

- ☐ know the metric prefixes: kilo, centi, milli
- ☐ multiply to convert to a smaller unit, divide to convert to a larger unit
- ☐ convert units of time

Key terms Write the definition as your teacher explains it.

Metric

Convert

Notes

Level 1 · Getting started

Easy · 28 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Convert. Bigger unit to smaller: multiply.

a Convert 3200 m to km.

b Convert 0.042 kg to g.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **28 QUESTIONS**

REMEMBER $\text{km} \rightarrow \text{m} \times 1000$, $\text{m} \rightarrow \text{cm} \times 100$, $\text{cm} \rightarrow \text{mm} \times 10$.

ROUND 1 Questions 1–12: convert. Big unit to small unit: multiply. Small to big: divide.

1 0.22 m to cm

2 0.5 cm to mm

3 36 000 mm to m

4 0.28 kL to L

5 48 m to mm

6 130 mL to L

7 23 m to mm

8 150 mm to m

9 450 L to kL

10 53 L to mL

11 3.3 g to mg

12 31 km to m

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 13–28: convert the time.

- | | | | | | | | |
|-------|--------------|-------|--------------|-------|-------------|-------|--------------|
| 13 | 3 min to s | 14 | 120 min to h | 15 | 2 h to min | 16 | 7 h to min |
| <hr/> | | <hr/> | | <hr/> | | <hr/> | |
| 17 | 5 h to min | 18 | 6 days to h | 19 | 10 min to s | 20 | 6 min to s |
| <hr/> | | <hr/> | | <hr/> | | <hr/> | |
| 21 | 660 min to h | 22 | 3 days to h | 23 | 4 days to h | 24 | 180 min to h |
| <hr/> | | <hr/> | | <hr/> | | <hr/> | |
| 25 | 8 min to s | 26 | 360 min to h | 27 | 7 min to s | 28 | 1 h to min |
| <hr/> | | <hr/> | | <hr/> | | <hr/> | |

 Level 2 · Building up

Medium · 22 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Convert capacity and time.

- a** Convert 8.3 L to mL.

Your teacher works this out. Copy every step.

- b** Convert 240 min to h.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

22 QUESTIONS

REMEMBER $\text{kg} \rightarrow \text{g} \times 1000$, $\text{t} \rightarrow \text{kg} \times 1000$, $\text{L} \rightarrow \text{mL} \times 1000$.

ROUND 1

Questions 1–16: convert.

- | | | | |
|--------------------------|---------------------------|----------------------------|-------------------------|
| 1 5.7 m to cm
_____ | 2 51 g to mg
_____ | 3 3 m to mm
_____ | 4 8.3 m to cm
_____ |
| 5 5 500 cm to m
_____ | 6 1.8 m to cm
_____ | 7 4 mm to cm
_____ | 8 56 kL to L
_____ |
| 9 1 800 m to km
_____ | 10 8 200 g to kg
_____ | 11 91 cm to mm
_____ | 12 1.7 km to m
_____ |
| 13 96 m to cm
_____ | 14 4 600 g to kg
_____ | 15 50 000 L to kL
_____ | 16 9.1 kg to g
_____ |

YOU DO

On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2

Questions 17–22: convert the time. Write your answer in the unit given.

17 4.75 h to min

18 9.25 h to min

19 525 min to h

20 255 min to h

21 570 min to h

22 1.75 days to h



Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Convert, then solve.

a Convert 15 h to seconds.

Your teacher works this out. Copy every step.

b How many minutes has Isla lived by her 12th birthday? (365 days a year)

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER

Bigger unit $\rightarrow$ smaller unit: multiply. Smaller $\rightarrow$ bigger: divide.

1 How many minutes has Isla lived by her 12th birthday? (365 days a year)

2 A recipe needs 350 g of flour for one cake. How many kg for 12 cakes?

3 A tap drips 5 mL a minute. How many litres in one day?

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name	Class
John	Math
Jane	Math
John	Science
Jane	Science
John	History
Jane	History
John	Art
Jane	Art
John	Music
Jane	Music
John	Physical Education
Jane	Physical Education
John	Language Arts
Jane	Language Arts
John	Math
Jane	Math
John	Science
Jane	Science
John	History
Jane	History
John	Art
Jane	Art
John	Music
Jane	Music
John	Physical Education
Jane	Physical Education
John	Language Arts
Jane	Language Arts



Convert 4.5 m to cm.



Convert 2750 g to kg.



Convert 3.5 h to minutes.

 Extension · Stretch yourself Worked example

Light travels 300 000 km in one second. How far does it travel in one hour?

- 1 hour = 3600 s
- $300\,000 \times 3600$

Answer: 1 080 000 000 km

YOU DO

On your own: read the worked example, then try these.

- E1 Your heart beats about 70 times a minute. About how many times does it beat in a year?

- E2 Convert 1 million seconds to days (1 d.p.).

Lesson summary

Missed the lesson? Start here.

Conversions 1. $\text{km} \rightarrow \text{m} \times 1000$, $\text{m} \rightarrow \text{cm} \times 100$, $\text{cm} \rightarrow \text{mm} \times 10$. 2. $\text{kg} \rightarrow \text{g} \times 1000$, $\text{t} \rightarrow \text{kg} \times 1000$, $\text{L} \rightarrow \text{mL} \times 1000$.
3. Bigger unit $\rightarrow$ smaller unit: multiply. Smaller $\rightarrow$ bigger: divide. 4. Time: $60 \text{ s} = 1 \text{ min}$, $60 \text{ min} = 1 \text{ h}$, $24 \text{ h} = 1 \text{ day}$.

 Level 1

3200 m to km

1. $\div 1000$

Answer: 3.2 km

 Level 2

8.3 L to mL

1. $\times 1000$

Answer: 8300 mL

 Level 3

15 h to s

1. $15 \times 60 \times 60$

Answer: 54 000 s

 Extension

Light in 1 hour

1. $300\,000 \times 3600$

Answer: 1 080 000 000 km

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LESSON
8.02

Perimeter

Date _____

Learning intentions

We are learning to:

- find the perimeter of shapes

Success criteria

I can:

- ☐ add all the side lengths
- ☐ find missing side lengths first
- ☐ use the same units for every side

Key terms Write the definition as your teacher explains it.

Perimeter

Composite shape

Notes

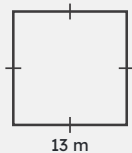
Level 1 · Getting started

Easy · 28 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find the perimeter.

a Find the perimeter of the square.



Your teacher works this out. Copy every step.

b Find the perimeter.



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **28 QUESTIONS**

REMEMBER Perimeter = distance around the outside.

ROUND 1 Questions 1–12: find the perimeter of the rectangle.

- | | | | |
|--------------------------------|--------------------------------|---------------------------------|--------------------------------|
| 1 10 m by 7 m
_____ | 2 29 m by 19 m
_____ | 3 3 m by 2 m
_____ | 4 6 m by 1 m
_____ |
| 5 16 m by 14 m
_____ | 6 14 m by 4 m
_____ | 7 23 m by 14 m
_____ | 8 2 m by 1 m
_____ |
| 9 23 m by 13 m
_____ | 10 6 m by 2 m
_____ | 11 14 m by 12 m
_____ | 12 20 m by 6 m
_____ |

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 13–28: find the perimeter of the regular shape.

13 regular octagon, side 5 cm

14 regular pentagon, side 17 cm

15 regular octagon, side 24 cm

16 square, side 13 cm

17 square, side 24 cm

18 regular hexagon, side 4 cm

19 regular hexagon, side 11 cm

20 regular hexagon, side 10 cm

21 regular octagon, side 16 cm

22 regular hexagon, side 8 cm

23 equilateral triangle, side 3 cm

24 regular octagon, side 6 cm

25 regular hexagon, side 17 cm

26 equilateral triangle, side 12 cm

27 regular octagon, side 11 cm

28 equilateral triangle, side 18 cm

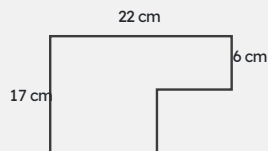
Level 2 · Building up

Medium · 27 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find the perimeter. Find missing sides first.

a Find the perimeter.



Your teacher works this out. Copy every step.

b Find the perimeter of a rectangle 2.4 m by 85 cm (in cm).

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

27 QUESTIONS

REMEMBER Add every side length.

ROUND 1 Questions 1–15: the perimeter and one side of a rectangle are given. Find the other side.

1 P = 76 m, length 25 m

2 P = 60 m, length 21 m

3 P = 76 m, length 34 m

4 P = 12 m, length 3 m

5 P = 16 m, length 5 m

6 P = 30 m, length 10 m

7 P = 14 m, length 4 m

8 P = 34 m, length 13 m

9 P = 26 m, length 8 m

10 P = 46 m, length 21 m

11 P = 42 m, length 11 m

12 P = 98 m, length 26 m

13 P = 88 m, length 30 m

14 P = 68 m, length 29 m

15 P = 96 m, length 26 m



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

A rectangle's length is 3 times its width. Its perimeter is 64 cm. Find its dimensions.

1. Width w , length $3w$
2. $2(w + 3w) = 8w = 64$
3. $w = 8$

Answer: 8 cm by 24 cm

YOU DO

On your own: read the worked example, then try these.

- E1** A rectangle's length is 5 cm more than its width. Its perimeter is 50 cm. Find its dimensions.

- E2** An isosceles triangle has perimeter 40 cm and base 12 cm. Find the equal sides.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the perimeter of a rectangle 12 cm by 7 cm.

Level 2

Find the perimeter of an equilateral triangle with sides 9.5 m.

Level 3

A square has perimeter 60 mm. Find its side.

Lesson summary

Missed the lesson? Start here.

Remember

1. Perimeter = distance around the outside.
2. Add every side length.
3. Find missing sides using opposite sides.
4. Check all units are the same.

Level 1

Square, side 13 m

1. 4×13

Answer: 52 m

Level 2

Rectangle 13 mm $\times$ 5 mm

1. $2 \times (13 + 5)$

Answer: 36 mm

Level 3

L-shape 22, 6, 17...

1. Missing sides: $22 - 10 = 12$, $17 - 6 = 11$

Answer: 78 cm

Extension

Length = 3 $\times$ width, P = 64

1. $8w = 64$

Answer: 8 cm by 24 cm

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
8.03

Circumference of a circle

Date _____

Learning intentions

We are learning to:

- find the circumference of a circle

Success criteria

I can:

- ☐ name the parts of a circle: centre, radius, diameter, circumference
- ☐ use $C = \pi d$ or $C = 2\pi r$
- ☐ round the answer to 1 decimal place

Key terms Write the definition as your teacher explains it.

Circumference

Radius

Diameter

Notes

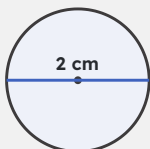
Level 1 · Getting started

Easy · 28 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find the circumference (1 decimal place).

a Find the circumference (1 d.p.).



b Find the circumference (1 d.p.).



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **28 QUESTIONS**

REMEMBER Diameter = $2 \times$ radius.

ROUND 1 Questions 1–12: find the circumference, $C = \pi d$. Round to 1 decimal place.

1 $d = 32$ cm

2 $d = 14$ cm

3 $d = 9$ cm

4 $d = 19$ cm

5 $d = 28$ cm

6 $d = 6$ cm

7 $d = 22$ cm

8 $d = 11$ cm

9 $d = 35$ cm

10 $d = 13$ cm

11 $d = 10$ cm

12 $d = 5$ cm

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 13–28: find the circumference, $C = 2\pi r$. Round to 1 decimal place.

13 $r = 6 \text{ m}$

14 $r = 20 \text{ m}$

15 $r = 15 \text{ m}$

16 $r = 16 \text{ m}$

17 $r = 5 \text{ m}$

18 $r = 8 \text{ m}$

19 $r = 2 \text{ m}$

20 $r = 9 \text{ m}$

21 $r = 3 \text{ m}$

22 $r = 12 \text{ m}$

23 $r = 14 \text{ m}$

24 $r = 7 \text{ m}$

25 $r = 11 \text{ m}$

26 $r = 1 \text{ m}$

27 $r = 21 \text{ m}$

28 $r = 13 \text{ m}$

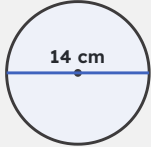
Level 2 · Building up

Medium · 24 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find C (1 decimal place).

a Find C (1 d.p.).



b Find C (1 d.p.).



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

24 QUESTIONS

REMEMBER $C = \pi d$ or $C = 2\pi r$.

ROUND 1 Questions 1–12: the circumference is given. Find the diameter, $d = C \div \pi$ (1 d.p.).

1 $C = 147$ cm

2 $C = 60$ cm

3 $C = 128$ cm

4 $C = 119$ cm

5 $C = 54$ cm

6 $C = 40$ cm

7 $C = 71$ cm

8 $C = 92$ cm

9 $C = 36$ cm

10 $C = 93$ cm

11 $C = 90$ cm

12 $C = 72$ cm

YOU DO

On your own: **Medium** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2

Questions 13–24: find the perimeter of the semicircle: half the circumference plus the diameter (1 d.p.).

13 $d = 20 \text{ cm}$

14 $d = 10 \text{ cm}$

15 $d = 17 \text{ cm}$

16 $d = 7 \text{ cm}$

17 $d = 11 \text{ cm}$

18 $d = 26 \text{ cm}$

19 $d = 16 \text{ cm}$

20 $d = 5 \text{ cm}$

21 $d = 12 \text{ cm}$

22 $d = 23 \text{ cm}$

23 $d = 15 \text{ cm}$

24 $d = 25 \text{ cm}$

Level 3 • Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Solve.

- a** A bike wheel has diameter 66 cm. How far does it travel in 100 turns (in m)?

- b** A circle has circumference 50 cm. Find its diameter (1 d.p.).

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER Use the π key on your calculator.

- 1** A bike wheel has diameter 66 cm. How far does it travel in 100 turns (in m, 1 d.p.)?

- 2 A circle has circumference 50 cm. Find its diameter (1 d.p.).

- 3 Find the perimeter of a semicircle with diameter 12 cm (1 d.p.).



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

Find the perimeter of a quarter circle with radius 10 cm (1 d.p.).

1. Arc = $\frac{1}{4} \times 2\pi \times 10 = 15.707\dots$
2. Add two radii: $15.71 + 20$

Answer: 35.7 cm

YOU DO

On your own: read the worked example, then try these.

- E1** Find the perimeter of a semicircle with radius 7 m (1 d.p.).

- E2** A running track has two straights of 100 m and two semicircles of diameter 64 m. Find its length (1 d.p.).

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

A circle has radius 8 cm. Find its diameter.

Level 2

Find the circumference of a circle with diameter 9 m (1 d.p.).

Level 3

Find the circumference of a circle with radius 3.5 cm (1 d.p.).

Lesson summary

Missed the lesson? Start here.

Remember

1. Diameter = $2 \times$ radius.
2. $C = \pi d$ or $C = 2\pi r$.
3. Use the π key on your calculator.
4. Round to 1 decimal place and include units.

Level 1

$$d = 2 \text{ cm}$$

$$1. \pi \times 2 = 6.283\dots$$

Answer: 6.3 cm

Level 2

$$r = 10 \text{ mm}$$

$$1. 2 \times \pi \times 10$$

Answer: 62.8 mm

Level 3

$$C = 50, \text{ find } d$$

$$1. 50 \div \pi$$

Answer: 15.9 cm

Extension

$$\text{Quarter circle } r = 10$$

$$1. 15.7 + 20$$

Answer: 35.7 cm

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
8.04

Area units

Date _____

Learning intentions

We are learning to:

- convert between units of area

Success criteria

I can:

- ☐ know $1 \text{ cm}^2 = 100 \text{ mm}^2$, $1 \text{ m}^2 = 10\,000 \text{ cm}^2$
- ☐ multiply to convert to a smaller unit, divide to a larger unit
- ☐ choose a sensible unit of area

Key terms Write the definition as your teacher explains it.

Area

Square unit

Notes

Level 1 · Getting started

Easy · 28 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Explain, then convert.**a** Why is $1 \text{ cm}^2 = 100 \text{ mm}^2$?

Your teacher works this out. Copy every step.

b How many cm^2 in 9.1 m^2 ?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **28 QUESTIONS****REMEMBER** $1 \text{ cm}^2 = 10 \times 10 = 100 \text{ mm}^2$.**ROUND 1** Questions 1–12: convert. $1 \text{ cm}^2 = 100 \text{ mm}^2$, $1 \text{ m}^2 = 10\,000 \text{ cm}^2$, $1 \text{ ha} = 10\,000 \text{ m}^2$.**1** 0.54 m^2 to cm^2

2 $4\,900 \text{ ha}$ to km^2

3 43 cm^2 to mm^2

4 29 km^2 to ha

5 $2\,700 \text{ mm}^2$ to cm^2

6 33 cm^2 to mm^2

7 0.26 km^2 to ha

8 1.2 m^2 to cm^2

9 4.1 m^2 to cm^2

10 31 cm^2 to mm^2

11 $6\,000 \text{ ha}$ to km^2

12 0.41 m^2 to cm^2

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 17–22: which unit would you use to measure the area? Write mm², cm², m², ha or km².

17 a classroom floor

18 a city

19 a country

20 a postcard

21 a mobile phone screen

22 a house block

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Convert.

a How many m² in 240 000 mm²?

b How many mm² in 8.6 m²?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Choose and compare.

3 QUESTIONS

REMEMBER 1 ha = 10 000 m²; 1 km² = 1 000 000 m².

1 Which unit would you use for: a stamp, a classroom floor, a farm, a country?

2 Which is larger: 3.2 m² or 30 000 cm²?

3 A tile is 400 cm². How many tiles cover 6 m²?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Convert 5 cm² to mm².

Level 2

Convert 30 000 cm² to m².

Level 3

Convert 1.5 m² to mm².

 Extension · Stretch yourself Worked example

Show that $1 \text{ km}^2 = 100 \text{ ha}$.

- $1 \text{ km}^2 = 1000 \text{ m} \times 1000 \text{ m} = 1\,000\,000 \text{ m}^2$
- $1 \text{ ha} = 10\,000 \text{ m}^2$
- $1\,000\,000 \div 10\,000$

Answer: 100 ha

YOU DO

On your own: read the worked example, then try these.

- E1 How many mm^2 are in 1 m^2 ?

- E2 A soccer field is 105 m by 68 m. What fraction of a hectare is it (2 d.p.)?

Lesson summary

Missed the lesson? Start here.

Area conversions

1. $1 \text{ cm}^2 = 10 \times 10 = 100 \text{ mm}^2$.

2. $1 \text{ m}^2 = 100 \times 100 = 10\,000 \text{ cm}^2$.

3. $1 \text{ ha} = 10\,000 \text{ m}^2$; $1 \text{ km}^2 = 1\,000\,000 \text{ m}^2$.

4. To a smaller unit: $\times$. To a larger unit: $\div$.

 Level 1

9.1 m^2 to cm^2

1. $\times 10\,000$

Answer: $91\,000 \text{ cm}^2$

 Level 2

240 mm^2 to cm^2

1. $\div 100$

Answer: 2.4 cm^2

 Level 3

$240\,000 \text{ mm}^2$ to m^2

1. $\div 1\,000\,000$

Answer: 0.24 m^2

 Extension

1 km^2 in ha

1. $1\,000\,000 \div 10\,000$

Answer: 100 ha

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LESSON
8.05

Area of squares and rectangles

Date _____

Learning intentions

We are learning to:

- find the area of squares and rectangles

Success criteria

I can:

- ☐ use $A = l \times w$ for a rectangle
- ☐ use $A = s^2$ for a square
- ☐ find a missing side from the area

Key terms Write the definition as your teacher explains it.

Length

Width

Notes

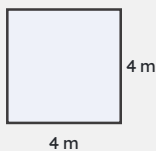
Level 1 · Getting started

Easy · 28 questions

WE DO With your teacher: copy what they write on the board.

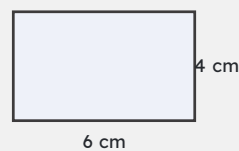
Example 1 Find the area.

a Find the area.



Your teacher works this out. Copy every step.

b Find the area.



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **28 QUESTIONS**

REMEMBER Area of a rectangle = length $\times$ width.

ROUND 1 Questions 1–12: find the area of the rectangle, $A = lw$.

- | | | | |
|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 16 m by 7 m
_____ | 2 4 m by 1 m
_____ | 3 8 m by 5 m
_____ | 4 17 m by 4 m
_____ |
| 5 16 m by 16 m
_____ | 6 17 m by 7 m
_____ | 7 20 m by 2 m
_____ | 8 14 m by 5 m
_____ |
| 9 3 m by 2 m
_____ | 10 18 m by 2 m
_____ | 11 18 m by 4 m
_____ | 12 11 m by 2 m
_____ |

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 13–28: find the area of the square, $A = s^2$.

13 side 19 cm

14 side 2 cm

15 side 7 cm

16 side 10 cm

17 side 15 cm

18 side 8 cm

19 side 1 cm

20 side 9 cm

21 side 18 cm

22 side 14 cm

23 side 6 cm

24 side 4 cm

25 side 13 cm

26 side 17 cm

27 side 5 cm

28 side 3 cm

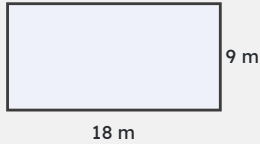
Level 2 · Building up

Medium · 27 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find the area. Use the same units.

a Find the area.



Your teacher works this out. Copy every step.

b A rectangle 2.5 m by 60 cm. Find its area in m^2 .

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line. **27 QUESTIONS**

REMEMBER Area of a square = side $\times$ side = s^2 .

ROUND 1 Questions 1–15: the area and one side of a rectangle are given. Find the other side.

1 $A = 32 \text{ m}^2$, width 4 m

2 $A = 60 \text{ m}^2$, width 4 m

3 $A = 128 \text{ m}^2$, width 8 m

4 $A = 22 \text{ m}^2$, width 11 m

5 $A = 54 \text{ m}^2$, width 9 m

6 $A = 266 \text{ m}^2$, width 14 m

7 $A = 52 \text{ m}^2$, width 4 m

8 $A = 42 \text{ m}^2$, width 3 m

9 $A = 168 \text{ m}^2$, width 12 m

10 $A = 72 \text{ m}^2$, width 4 m

11 $A = 12 \text{ m}^2$, width 4 m

12 $A = 25 \text{ m}^2$, width 5 m

13 $A = 150 \text{ m}^2$, width 10 m

14 $A = 20 \text{ m}^2$, width 2 m

15 $A = 27 \text{ m}^2$, width 3 m

YOU DO On your own: **Medium** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2 Questions 16–27: find the area.

16 6 cm by 6 cm

17 1.9 cm by 12 cm

18 1.8 cm by 12 cm

19 4.7 cm by 6 cm

20 1.9 cm by 6 cm

21 9 cm by 3 cm

22 5.9 cm by 6 cm

23 6.2 cm by 10 cm

24 4.5 cm by 9 cm

25 7.2 cm by 9 cm

26 8.1 cm by 11 cm

27 6.5 cm by 7 cm

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find the missing side.

a A square has area 81 cm^2 . Find its side.

Your teacher works this out. Copy every step.

b A rectangle has area 48 m^2 and length 8 m. Find its width.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER Use the same units for both sides.

1 Carpet costs \$45 per m^2 . Find the cost to carpet a room 5.2 m by 4 m.

2 How many 30 cm by 30 cm tiles cover a floor 3.6 m by 2.7 m?

3 A rectangle has perimeter 26 cm and area 40 cm^2 . Find its length and width.

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

A 20 m by 15 m lawn has a 2 m wide path around the outside of it. Find the area of the path.

1. Outer rectangle: 24 m by 19 m = 456 m²
2. Lawn: 20 × 15 = 300 m²
3. 456 – 300

Answer: 156 m²

YOU DO

On your own: read the worked example, then try these.

- E1** A 6 m by 4 m pool has a 1 m wide path around it. Find the path area.

- E2** Which rectangle with perimeter 24 cm has the largest area?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the area of a rectangle 9 cm by 6 cm.

Level 2

Find the area of a square with side 2.5 m.

Level 3

A rectangle has area 60 m² and width 5 m. Find its length.

Lesson summary

Missed the lesson? Start here.

Remember

1. Area of a rectangle = length $\times$ width.
3. Use the same units for both sides.

2. Area of a square = side $\times$ side = s^2 .
4. Answer in square units.

Level 1

4 m square

1. 4×4

Answer: 16 m^2

Level 2

$18 \text{ m} \times 9 \text{ m}$

1. 18×9

Answer: 162 m^2

Level 3

Area 48, length 8

1. $48 \div 8$

Answer: 6 m

Extension

Path around a lawn

1. $456 - 300$

Answer: 156 m^2

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
8.06

Area of a triangle

Date _____

Learning intentions

We are learning to:

- find the area of a triangle

Success criteria

I can:

- ☐ identify the base and the perpendicular height
- ☐ use $A = \frac{1}{2} \times b \times h$
- ☐ find the height when it is outside the triangle

Key terms Write the definition as your teacher explains it.

Base

Perpendicular height

Notes

Level 1 · Getting started

Easy · 25 questions

WE DO With your teacher: copy what they write on the board.

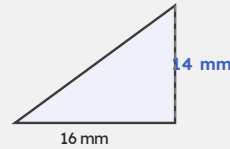
Example 1 Find the area. $A = \frac{1}{2}bh$.

a Find the area.



Your teacher works this out. Copy every step.

b Find the area.



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

25 QUESTIONS

REMEMBER $A = \frac{1}{2} \times \text{base} \times \text{height}$.

ROUND 1 Questions 1–9: find the area of the triangle, $A = \frac{1}{2} \times b \times h$.

1 $b = 4 \text{ cm}, h = 14 \text{ cm}$

2 $b = 18 \text{ cm}, h = 6 \text{ cm}$

3 $b = 4 \text{ cm}, h = 7 \text{ cm}$

4 $b = 16 \text{ cm}, h = 19 \text{ cm}$

5 $b = 2 \text{ cm}, h = 10 \text{ cm}$

6 $b = 6 \text{ cm}, h = 12 \text{ cm}$

7 $b = 14 \text{ cm}, h = 2 \text{ cm}$

8 $b = 6 \text{ cm}, h = 11 \text{ cm}$

9 $b = 20 \text{ cm}, h = 7 \text{ cm}$

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 10–25: find the area of the triangle.

10 $b = 16 \text{ m}, h = 19 \text{ m}$

11 $b = 15 \text{ m}, h = 20 \text{ m}$

12 $b = 9 \text{ m}, h = 11 \text{ m}$

13 $b = 22 \text{ m}, h = 18 \text{ m}$

14 $b = 8 \text{ m}, h = 25 \text{ m}$

15 $b = 14 \text{ m}, h = 4 \text{ m}$

16 $b = 23 \text{ m}, h = 22 \text{ m}$

17 $b = 11 \text{ m}, h = 25 \text{ m}$

18 $b = 22 \text{ m}, h = 5 \text{ m}$

19 $b = 23 \text{ m}, h = 16 \text{ m}$

20 $b = 24 \text{ m}, h = 21 \text{ m}$

21 $b = 25 \text{ m}, h = 19 \text{ m}$

22 $b = 17 \text{ m}, h = 20 \text{ m}$

23 $b = 20 \text{ m}, h = 9 \text{ m}$

24 $b = 19 \text{ m}, h = 25 \text{ m}$

25 $b = 10 \text{ m}, h = 13 \text{ m}$

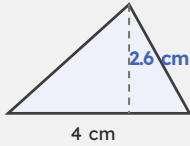
Level 2 · Building up

Medium · 24 questions

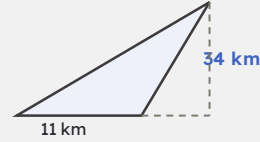
WE DO With your teacher: copy what they write on the board.

Example 2 Find the area. The height may be outside.

a Find the area.



b Find the area.



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

24 QUESTIONS

REMEMBER The height is perpendicular (at 90°) to the base.

ROUND 1 Questions 1–12: the area and the base are given. Find the height, $h = 2A \div b$.

1 $A = 48 \text{ cm}^2$, $b = 12 \text{ cm}$

2 $A = 72 \text{ cm}^2$, $b = 9 \text{ cm}$

3 $A = 52 \text{ cm}^2$, $b = 8 \text{ cm}$

4 $A = 30 \text{ cm}^2$, $b = 20 \text{ cm}$

5 $A = 44 \text{ cm}^2$, $b = 11 \text{ cm}$

6 $A = 17 \text{ cm}^2$, $b = 2 \text{ cm}$

7 $A = 150 \text{ cm}^2$, $b = 15 \text{ cm}$

8 $A = 28 \text{ cm}^2$, $b = 14 \text{ cm}$

9 $A = 33 \text{ cm}^2$, $b = 6 \text{ cm}$

10 $A = 91 \text{ cm}^2$, $b = 13 \text{ cm}$

11 $A = 26 \text{ cm}^2$, $b = 13 \text{ cm}$

12 $A = 16 \text{ cm}^2$, $b = 16 \text{ cm}$

YOU DO

On your own: **Medium** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2

Questions 13–24: find the area of the triangle.

13 $b = 1.1 \text{ m}, h = 2 \text{ m}$

14 $b = 6.7 \text{ m}$, $h = 8 \text{ m}$

15 $b = 7.4 \text{ m}, h = 10 \text{ m}$

16 $b = 8.2 \text{ m}, h = 8 \text{ m}$

17 $b = 2.3 \text{ m}, h = 12 \text{ m}$

18 $b = 1.3 \text{ m}, h = 8 \text{ m}$

19 $b = 1.6 \text{ m}$, $h = 9 \text{ m}$

20 $b = 4.4 \text{ m}, h = 8 \text{ m}$

21 $b = 6.4 \text{ m}, h = 11 \text{ m}$

22 $b = 2.8 \text{ m}, h = 2 \text{ m}$

23 $b = 3.6 \text{ m}, h = 6 \text{ m}$

24 $b = 8.6 \text{ m}, h = 3 \text{ m}$

Level 3 • Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Find the height, then explain.

a A triangle has area 36 cm^2 and base 9 cm . Find its height.

b Why is the area of a triangle half of a rectangle?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER The height may be outside the triangle.

1 A triangular sail has base 3.2 m and height 5 m. Find its area.

2 A triangular garden bed has base 6 m and height 4 m. Mulch covers 3 m^2 per bag. How many bags?

3 Find the area of a square of side 10 cm with one corner triangle (legs 4 cm) cut off.



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

Find the area of a kite with diagonals 10 cm and 16 cm.

1. The long diagonal splits it into 2 triangles, base 16, height 5
2. $2 \times \frac{1}{2} \times 16 \times 5$

Answer: 80 cm²

YOU DO

On your own: read the worked example, then try these.

- E1** Find the area of a rhombus with diagonals 6 m and 9 m.

- E2** A triangle has vertices (0, 0), (8, 0) and (3, 6) on a grid. Find its area.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the area: base 10 m, height 7 m.

Level 2

Find the area: base 5 cm, height 3.2 cm.

Level 3

A triangle has area 24 cm² and base 6 cm. Find its height.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. $A = \frac{1}{2} \times \text{base} \times \text{height}$.
 2. The height is perpendicular (at 90°) to the base.
 3. The height may be outside the triangle.
 4. Answer in square units.

Level 1

base 30, height 24

1. $\frac{1}{2} \times 30 \times 24$

Answer: 360 mm^2

Level 2

base 4, height 2.6

1. $\frac{1}{2} \times 4 \times 2.6$

Answer: 5.2 cm^2

Level 3

Area 36, base 9

1. $\frac{1}{2} \times 9 \times h = 36$

Answer: $h = 8 \text{ cm}$

Extension

Kite 10 by 16

1. $2 \times \frac{1}{2} \times 16 \times 5$

Answer: 80 cm^2

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
8.07

Area of a parallelogram

Date _____

Learning intentions

We are learning to:

- find the area of a parallelogram

Success criteria

I can:

- ☐ identify the base and perpendicular height
- ☐ use $A = b \times h$
- ☐ not use the slanted side as the height

Key terms Write the definition as your teacher explains it.

Parallelogram

Perpendicular height

Notes

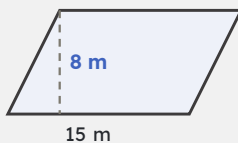
Level 1 · Getting started

Easy · 25 questions

WE DO With your teacher: copy what they write on the board.

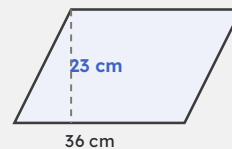
Example 1 Find the area. $A = bh$.

a Find the area.



Your teacher works this out. Copy every step.

b Find the area.



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **25 QUESTIONS**

REMEMBER $A = \text{base} \times \text{height}$.

ROUND 1 Questions 1–9: find the area of the parallelogram, $A = bh$.

1 $b = 8 \text{ m}, h = 15 \text{ m}$

2 $b = 5 \text{ m}, h = 6 \text{ m}$

3 $b = 18 \text{ m}, h = 3 \text{ m}$

4 $b = 5 \text{ m}, h = 14 \text{ m}$

5 $b = 6 \text{ m}, h = 3 \text{ m}$

6 $b = 4 \text{ m}, h = 3 \text{ m}$

7 $b = 17 \text{ m}, h = 2 \text{ m}$

8 $b = 14 \text{ m}, h = 9 \text{ m}$

9 $b = 16 \text{ m}, h = 9 \text{ m}$

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 10–25: find the area of the parallelogram.

10 $b = 22 \text{ cm}, h = 6 \text{ cm}$

11 $b = 35 \text{ cm}, h = 7 \text{ cm}$

12 $b = 29 \text{ cm}, h = 12 \text{ cm}$

13 $b = 30 \text{ cm}, h = 22 \text{ cm}$

14 $b = 36 \text{ cm}, h = 17 \text{ cm}$

15 $b = 25 \text{ cm}, h = 22 \text{ cm}$

16 $b = 36 \text{ cm}, h = 9 \text{ cm}$

17 $b = 31 \text{ cm}, h = 12 \text{ cm}$

18 $b = 13 \text{ cm}, h = 9 \text{ cm}$

19 $b = 18 \text{ cm}, h = 5 \text{ cm}$

20 $b = 24 \text{ cm}, h = 8 \text{ cm}$

21 $b = 13 \text{ cm}, h = 10 \text{ cm}$

22 $b = 27 \text{ cm}, h = 7 \text{ cm}$

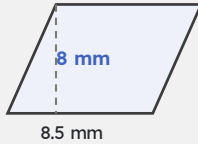
23 $b = 10 \text{ cm}, h = 18 \text{ cm}$

24 $b = 13 \text{ cm}, h = 5 \text{ cm}$

25 $b = 27 \text{ cm}, h = 17 \text{ cm}$

Level 2 · Building up

Medium · 24 questions

WE DO With your teacher: copy what they write on the board.**Example 2** Find the area, then explain.**a** Find the area.

Your teacher works this out. Copy every step.

b Why is the area of a parallelogram the same as a rectangle with the same base and height?

Your teacher works this out. Copy every step.

YOU DOOn your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

24 QUESTIONS

REMEMBER Height is perpendicular to the base.**ROUND 1** Questions 1–12: the area and one measurement are given. Find the other.**1** $A = 90 \text{ m}^2$, $b = 9 \text{ m}$. Find h .

2 $A = 144 \text{ m}^2$, $h = 12 \text{ m}$. Find b .

3 $A = 70 \text{ m}^2$, $b = 7 \text{ m}$. Find h .

4 $A = 96 \text{ m}^2$, $b = 16 \text{ m}$. Find h .

5 $A = 96 \text{ m}^2$, $h = 12 \text{ m}$. Find b .

6 $A = 204 \text{ m}^2$, $h = 12 \text{ m}$. Find b .

7 $A = 112 \text{ m}^2$, $h = 8 \text{ m}$. Find b .

8 $A = 95 \text{ m}^2$, $b = 19 \text{ m}$. Find h .

9 $A = 12 \text{ m}^2$, $b = 2 \text{ m}$. Find h .

10 $A = 22 \text{ m}^2$, $b = 11 \text{ m}$. Find h .

11 $A = 32 \text{ m}^2$, $h = 8 \text{ m}$. Find b .

12 $A = 240 \text{ m}^2$, $b = 20 \text{ m}$. Find h .

YOU DO

On your own: **Medium** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2

Questions 13–24: find the area.

13 $b = 4.8 \text{ m}, h = 2.8 \text{ m}$

14 $b = 2.5 \text{ m}, h = 4.7 \text{ m}$

15 $b = 9.1 \text{ m}, h = 1.6 \text{ m}$

16 $b = 6.9 \text{ m}$, $h = 2.2 \text{ m}$

17 $b = 1.8 \text{ m}$, $h = 2.9 \text{ m}$

18 $b = 8.7 \text{ m}$, $h = 4.1 \text{ m}$

19 $b = 7.1 \text{ m}, h = 4.9 \text{ m}$

20 $b = 3.1 \text{ m}, h = 3.4 \text{ m}$

21 $b = 1.7 \text{ m}, h = 4.8 \text{ m}$

22 $b = 7 \text{ m}, h = 4 \text{ m}$

23 $b = 7.6 \text{ m}, h = 4 \text{ m}$

24 $b = 5.9 \text{ m}, h = 2.1 \text{ m}$

Level 3 • Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Find the height, then the area.

- a** A parallelogram has area 72 cm^2 and base 12 cm . Find its height.

- b** A parallelogram has sides 10 cm and 6 cm, and height 5 cm on the 10 cm side. Find its area.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER Do not use the slanted side.

- 1** A parallelogram-shaped garden has base 12 m and height 7.5 m. Turf costs \$9.80 per m^2 . Find the cost.

- 2 Draw two different parallelograms with area 24 cm^2 .

- 3 A rhombus has side 10 cm and height 8 cm. Find its area.



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

Find the area of a trapezium with parallel sides 8 cm and 14 cm and height 5 cm.

- Two copies make a parallelogram with base $8 + 14 = 22$
- Half of 22×5

Answer: 55 cm^2

YOU DO

On your own: read the worked example, then try these.

- E1 Find the area of a trapezium with parallel sides 6 m and 10 m and height 4 m.

- E2 A trapezium has area 60 cm^2 , height 6 cm and one parallel side 8 cm. Find the other.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the area: base 9 cm, height 4 cm.

Level 2

Find the area: base 2.5 m, height 1.6 m.

Level 3

A parallelogram has area 50 m^2 and height 4 m. Find its base.

Lesson summary

Missed the lesson? Start here.

Remember

1. $A = \text{base} \times \text{height}$.
2. Height is perpendicular to the base.
3. Do not use the slanted side.
4. Cut the triangle off one end and move it: a rectangle.

Level 1

base 15, height 8

1. 15×8

Answer: 120 m^2

Level 2

base 8.5, height 8

1. 8.5×8

Answer: 68 mm^2

Level 3

Area 72, base 12

1. $72 \div 12$

Answer: 6 cm

Extension

Trapezium 8, 14, h 5

1. $\frac{1}{2} \times 22 \times 5$

Answer: 55 cm^2

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
8.08

Area of composite shapes

Date _____

Learning intentions

We are learning to:

- find the area of composite and shaded shapes

Success criteria

I can:

- ☐ split a shape into rectangles and triangles, then add
- ☐ find a shaded area by subtracting
- ☐ solve practical problems such as painting a wall

Key terms Write the definition as your teacher explains it.

Composite shape

Shaded area

Notes

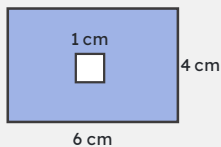
Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

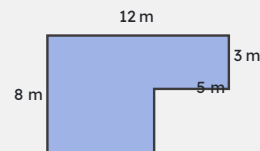
Example 1 Find the area. Add or subtract rectangles.

a Find the shaded area.



Your teacher works this out. Copy every step.

b Find the area.



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. 20 QUESTIONS

REMEMBER Split the shape into rectangles and triangles.

ROUND 1 Questions 1–6: find the area of the rectangle with a rectangular hole cut out (subtract).

① 9 cm × 12 cm rectangle, 4 cm × 3 cm hole

② 15 cm × 10 cm rectangle, 3 cm × 5 cm hole

③ 9 cm × 13 cm rectangle, 4 cm × 4 cm hole

④ 15 cm × 6 cm rectangle, 5 cm × 4 cm hole

⑤ 18 cm × 10 cm rectangle, 2 cm × 5 cm hole

⑥ 11 cm × 11 cm rectangle, 5 cm × 2 cm hole

YOU DO

On your own: **Easy** keep going: Round 2.

14 QUESTIONS MORE

ROUND 2

Questions 7–20: the shape is made of two rectangles. Find the total area (add).

7 $6\text{ m} \times 3\text{ m}$ and $6\text{ m} \times 8\text{ m}$

8 $3\text{ m} \times 7\text{ m}$ and $6\text{ m} \times 5\text{ m}$

9 $7\text{ m} \times 9\text{ m}$ and $9\text{ m} \times 3\text{ m}$

10 $12\text{ m} \times 2\text{ m}$ and $7\text{ m} \times 2\text{ m}$

11 $12\text{ m} \times 9\text{ m}$ and $6\text{ m} \times 3\text{ m}$

12 $8\text{ m} \times 2\text{ m}$ and $2\text{ m} \times 2\text{ m}$

13 $6\text{ m} \times 3\text{ m}$ and $8\text{ m} \times 7\text{ m}$

14 $9\text{ m} \times 4\text{ m}$ and $5\text{ m} \times 4\text{ m}$

15 $10\text{ m} \times 7\text{ m}$ and $6\text{ m} \times 4\text{ m}$

16 $10\text{ m} \times 3\text{ m}$ and $8\text{ m} \times 4\text{ m}$

17 $7\text{ m} \times 7\text{ m}$ and $5\text{ m} \times 2\text{ m}$

18 $6\text{ m} \times 8\text{ m}$ and $3\text{ m} \times 5\text{ m}$

19 $9\text{ m} \times 3\text{ m}$ and $6\text{ m} \times 6\text{ m}$

20 $10\text{ m} \times 6\text{ m}$ and $8\text{ m} \times 4\text{ m}$

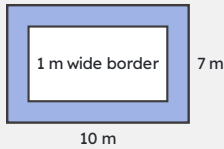
Level 2 · Building up

Medium · 9 questions

WE DO With your teacher: copy what they write on the board.

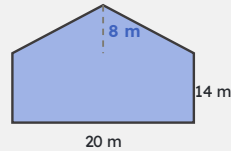
Example 2 Find the area.

a Find the shaded area.



Your teacher works this out. Copy every step.

b Find the area.



Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line (show working in the boxes).

9 QUESTIONS

REMEMBER Find missing lengths first.

ROUND 1 Questions 1–6: find the area: a rectangle with a triangle on top.

1 rectangle $5\text{ m} \times 3\text{ m}$, triangle base 5 m , height 6 m

2 rectangle $8\text{ m} \times 7\text{ m}$, triangle base 8 m , height 3 m

3 rectangle $11\text{ m} \times 7\text{ m}$, triangle base 11 m , height 4 m

4 rectangle $11\text{ m} \times 6\text{ m}$, triangle base 11 m , height 8 m

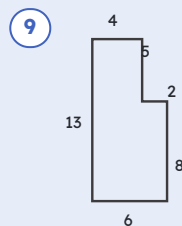
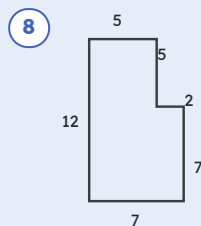
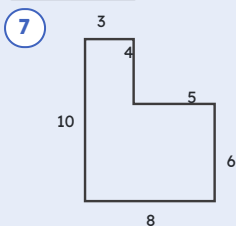
5 rectangle $5\text{ m} \times 10\text{ m}$, triangle base 5 m , height 6 m

6 rectangle $10\text{ m} \times 10\text{ m}$, triangle base 10 m , height 2 m

YOU DO On your own: **Medium** keep going: Round 2.

3 QUESTIONS MORE

ROUND 2 Questions 7–9: find the area of the L-shape. Split it into two rectangles.



Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve the problem.

- a** A wall 6.6 m by 3 m has a 2 m by 1 m window and a 0.9 m by 2 m door. One litre of paint covers 3 m². How many whole litres are needed?

Your teacher works this out. Copy every step.

- b** Find the area of a square of side 5 mm with a triangle of base 5 mm and height 5 mm on top.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER Add the parts, or subtract holes (shaded area).

- 1** A wall 6.6 m by 3 m has a 2 m by 1 m window and a 0.9 m by 2 m door. One litre of paint covers 3 m². How many whole litres are needed?

- 2** A 50 cm by 35 cm poster has a 40 cm by 28 cm photo on it. Find the area of the poster not covered.

- 3** Find the area of an L-shaped room: 6 m by 4 m with a 2 m by 1.5 m corner cut out.

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

A path 1 m wide runs along two sides of a 12 m by 8 m lawn (an L-shape). Find the area of the path.

1. Outer rectangle $13 \times 9 = 117 \text{ m}^2$
2. Lawn $12 \times 8 = 96 \text{ m}^2$
3. $117 - 96$

Answer: 21 m^2

YOU DO

On your own: read the worked example, then try these.

- E1** A 1.5 m wide path goes around all four sides of a 10 m by 6 m pool. Find its area.

- E2** Find the area of a cross made of five 3 cm squares.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the area of a 10 cm × 5 cm rectangle with a 2 cm × 2 cm square hole.

Level 2

Find the area of an L-shape made from a 6 m × 3 m and a 2 m × 2 m rectangle.

Level 3

A wall is 4 m × 2.5 m with a 1 m × 1 m window. Paint covers 3 m² per litre. How many whole litres?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Split the shape into rectangles and triangles.
 2. Find missing lengths first.
 3. Add the parts, or subtract holes (shaded area).
 4. Check units and give square units.

Level 1

6×4 rectangle with 1×1 hole

1. $24 - 1$

Answer: 23 cm^2

Level 2

L-shape 12×8 minus 5×5

1. $96 - 25$

Answer: 71 m^2

Level 3

Border 1 m wide on 10×7

1. $70 - 8 \times 5$

Answer: 30 m^2

Extension

Paint the wall

1. $19.8 - 2 - 1.8 = 16 \text{ m}^2$

2. $16 \div 3 = 5.3$

Answer: 6 L

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
8.09

Prisms, cross-sections and views

Date _____

Learning intentions

We are learning to:

- identify prisms and their cross-sections
- draw the front, side and top views of a solid

Success criteria

I can:

- ☐ know a prism has the same cross-section all the way through
- ☐ draw a cross-section parallel to the ends
- ☐ sketch front, side and top views

Key terms Write the definition as your teacher explains it.

Prism

Cross-section

View

Notes

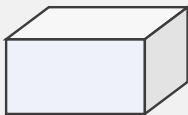
Level 1 · Getting started

Easy · 17 questions

WE DO With your teacher: copy what they write on the board.

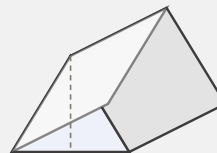
Example 1 Is it a prism? Draw its cross-section.

a Is it a prism? Draw its cross-section.



Your teacher works this out. Copy every step.

b Is it a prism? Draw its cross-section.



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **17 QUESTIONS**

REMEMBER A **PRISM** has two identical, parallel ends joined by rectangles.

ROUND 1 Questions 1–9: name the shape of the cross-section.

1 a can of soup

2 a pencil (six-sided)

3 hexagonal prism

4 octagonal prism

5 cylinder

6 a cereal box

7 cube

8 rectangular prism

9 pentagonal prism

YOU DO On your own: **Easy** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 10–17: how many faces (F), vertices (V) and edges (E)?

10 octagonal prism

11 hexagonal prism

12 rectangular prism

13 pentagonal prism

14 square pyramid

15 cube

16 triangular pyramid

17 triangular prism

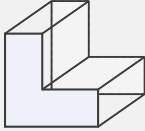
Level 2 · Building up

Medium · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Draw the cross-section, then explain.

a Draw the cross-section.



Your teacher works this out. Copy every step.

b Is a cone a prism? Explain.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

12 QUESTIONS

REMEMBER The **CROSS-SECTION** is the same shape all the way through.

ROUND 1 Questions 1–12: is it a prism? Write yes or no.

1 cube

2 cone

3 a tent with triangle ends

4 cylinder

5 sphere

6 a party hat

7 triangular prism

8 rectangular prism

9 hexagonal prism

10 square pyramid

11 a book

12 a brick

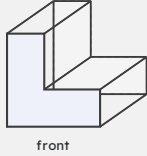
Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

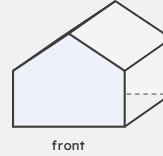
Example 3 Sketch the three views.

a Sketch the front, left and top views.



Your teacher works this out. Copy every step.

b Sketch the right, front and top views.



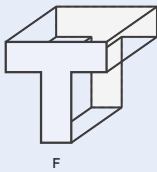
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Views and solids.

2 QUESTIONS

REMEMBER Views: look from the front, the side and the top.

1 Sketch the front, side and top views.



2 A solid has a square front view, a square side view and a circle top view. What is it?

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

A solid has front view a rectangle, side view a triangle and top view a rectangle. What is it?

1. A triangle from the side means triangular ends
2. Rectangles from the front and top: it is lying on a rectangular face

Answer: a triangular prism

YOU DO

On your own: read the worked example, then try these.

- E1 A solid has front and side views both triangles and top view a square. What is it?

- E2 Draw the front view of a stack of cubes: 3 in the back row, 2 in the middle, 1 at the front, all in a line.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Is a cylinder a prism? Explain.

Level 2

Name the cross-section of a hexagonal prism.

Level 3

Sketch the top view of a triangular prism lying on its rectangular face.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. A **prism** has two identical, parallel ends joined by rectangles.
 2. The **cross-section** is the same shape all the way through.
 3. Views: look from the front, the side and the top.
 4. Pyramids, cones and spheres are not prisms.

Level 1

Rectangular box

1. Same rectangle all through

Answer: prism,
rectangle cross-section

Level 2

Cone

1. Cross-section shrinks

Answer: not a prism

Level 3

L-shaped prism

1. Front: an L

Answer: L-shape,
rectangle, rectangle

Extension

Rectangle, triangle,
rectangle

1. Triangular ends

Answer: triangular
prism

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
8.10

Volume units

Date _____

Learning intentions

We are learning to:

- convert between units of volume

Success criteria

I can:

- ☐ know $1 \text{ cm}^3 = 1000 \text{ mm}^3$ and $1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$
- ☐ multiply to a smaller unit, divide to a larger unit
- ☐ count cubes to find volume

Key terms Write the definition as your teacher explains it.

Volume

Cubic unit

Notes

Level 1 · Getting started

Easy · 28 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Explain, then convert.

a Why is $1 \text{ cm}^3 = 1000 \text{ mm}^3$?

b $20 \text{ cm}^3 = ___ \text{ mm}^3$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

28 QUESTIONS

REMEMBER $1 \text{ cm}^3 = 10 \times 10 \times 10 = 1000 \text{ mm}^3$.

ROUND 1 Questions 1–12: convert. $1 \text{ cm}^3 = 1000 \text{ mm}^3$, $1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$.

1 27 cm^3 to mm^3

2 38 m^3 to cm^3

3 160 mm^3 to cm^3

4 $2\,300\,000 \text{ cm}^3$ to m^3

5 9 cm^3 to mm^3

6 $420\,000 \text{ cm}^3$ to m^3

7 $14\,000 \text{ mm}^3$ to cm^3

8 37 cm^3 to mm^3

9 53 m^3 to cm^3

10 $3\,000 \text{ mm}^3$ to cm^3

11 0.54 m^3 to cm^3

12 $6\,000 \text{ mm}^3$ to cm^3

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 17–22: convert. $1 \text{ m}^3 = 1000 \text{ L} = 1 \text{ kL}$.

17 43 m^3 to L

18 61 m^3 to L

19 3 m^3 to L

20 15 000 L to m^3

21 47 000 L to m^3

22 77 000 L to m^3

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Convert, then count cubes.

a $230\,000 \text{ mm}^3 = \text{---} \text{ m}^3$

Your teacher works this out. Copy every step.

b How many 1 cm cubes fit in a 5 cm by 4 cm by 3 cm box?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Cubes and volume.

3 QUESTIONS

REMEMBER To a smaller unit: $\times$. To a larger unit: $\div$.

1 How many 1 cm cubes fit in a box 5 cm by 4 cm by 3 cm?

2 How many 2 cm cubes fit in a box 10 cm by 6 cm by 4 cm?

3 Order: 0.02 m^3 , $15\,000 \text{ cm}^3$, $25\,000\,000 \text{ mm}^3$.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Convert 6 cm^3 to mm^3 .

Level 2

Convert $2\,500\,000 \text{ cm}^3$ to m^3 .

Level 3

Convert 1500 mm^3 to cm^3 .

Extension · Stretch yourself**Worked example**How many mm^3 in 1 m^3 ?

1. $1 \text{ m} = 1000 \text{ mm}$
2. $1000 \times 1000 \times 1000$

Answer: 1 000 000 000 mm^3 **YOU DO**

On your own: read the worked example, then try these.

- E1** Convert $0.000\,35 \text{ m}^3$ to mm^3 .

- E2** A cube has volume 1 m^3 . How many 10 cm cubes fit inside it?

Lesson summary

Missed the lesson? Start here.

- Volume conversions**
1. $1 \text{ cm}^3 = 10 \times 10 \times 10 = 1000 \text{ mm}^3$.
 2. $1 \text{ m}^3 = 100 \times 100 \times 100 = 1\,000\,000 \text{ cm}^3$.
 3. To a smaller unit: $\times$. To a larger unit: $\div$.
 4. Volume is measured in cubic units.

Level 120 cm^3 to mm^3

1. $\times 1000$

Answer: 20 000 mm^3 **Level 2**0.5 m^3 to cm^3

1. $\times 1\,000\,000$

Answer: 500 000 cm^3 **Level 3**230 000 mm^3 to m^3

1. $\div 1\,000\,000\,000$

Answer: 0.00023 m^3 **Extension**1 m^3 in mm^3

1. 1000^3

Answer: 1 000 000 000

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
8.11

Volume of a rectangular prism

Date _____

Learning intentions

We are learning to:

- find the volume of rectangular prisms

Success criteria

I can:

- ☐ use $V = l \times w \times h$
- ☐ use $V = \text{area of base} \times \text{height}$
- ☐ find a missing dimension

Key terms Write the definition as your teacher explains it.

Volume

Base area

Notes

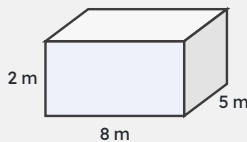
Level 1 · Getting started

Easy · 25 questions

WE DO With your teacher: copy what they write on the board.

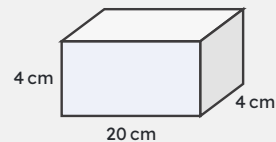
Example 1 Find the volume. $V = lwh$.

a Find the volume.



Your teacher works this out. Copy every step.

b Find the volume.



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **25 QUESTIONS**

REMEMBER $V = \text{length} \times \text{width} \times \text{height}$.

ROUND 1 Questions 1–9: find the volume of the rectangular prism, $V = lwh$.

1 $5 \times 6 \times 2 \text{ cm}$

2 $12 \times 6 \times 4 \text{ cm}$

3 $11 \times 2 \times 9 \text{ cm}$

4 $9 \times 3 \times 4 \text{ cm}$

5 $12 \times 9 \times 1 \text{ cm}$

6 $10 \times 9 \times 7 \text{ cm}$

7 $6 \times 3 \times 9 \text{ cm}$

8 $6 \times 7 \times 6 \text{ cm}$

9 $2 \times 8 \times 2 \text{ cm}$

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 10–25: find the volume of the cube, $V = s^3$.

10 side 12 cm

11 side 1 cm

12 side 6 cm

13 side 9 m

14 side 11 cm

15 side 5 m

16 side 7 cm

17 side 10 m

18 side 5 cm

19 side 6 m

20 side 2 cm

21 side 8 m

22 side 10 cm

23 side 4 cm

24 side 2 m

25 side 8 cm

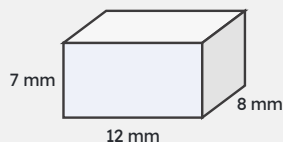
Level 2 · Building up

Medium · 24 questions

WE DO With your teacher: copy what they write on the board.

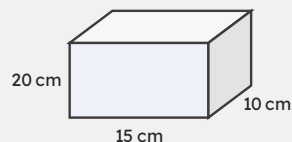
Example 2 Find the volume.

a Find the volume.



Your teacher works this out. Copy every step.

b Find the volume.



Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

24 QUESTIONS

REMEMBER Or $V = \text{area of base} \times \text{height}$.

ROUND 1 Questions 1–12: the volume and two measurements are given. Find the third.

1 $V = 70 \text{ cm}^3$, $l = 7 \text{ cm}$, $w = 2 \text{ cm}$.
Find h .

2 $V = 432 \text{ cm}^3$, $l = 12 \text{ cm}$, $w = 6 \text{ cm}$.
Find h .

3 $V = 54 \text{ cm}^3$, $l = 6 \text{ cm}$, $w = 3 \text{ cm}$.
Find h .

4 $V = 84 \text{ cm}^3$, $l = 3 \text{ cm}$, $w = 4 \text{ cm}$.
Find h .

5 $V = 84 \text{ cm}^3$, $l = 2 \text{ cm}$, $w = 6 \text{ cm}$.
Find h .

6 $V = 960 \text{ cm}^3$, $l = 12 \text{ cm}$, $w = 10 \text{ cm}$.
Find h .

7 $V = 352 \text{ cm}^3$, $l = 11 \text{ cm}$, $w = 8 \text{ cm}$.
Find h .

8 $V = 300 \text{ cm}^3$, $l = 5 \text{ cm}$, $w = 10 \text{ cm}$.
Find h .

9 $V = 36 \text{ cm}^3$, $l = 2 \text{ cm}$, $w = 2 \text{ cm}$.
Find h .

10 $V = 70 \text{ cm}^3$, $l = 5 \text{ cm}$, $w = 2 \text{ cm}$.
Find h .

11 $V = 392 \text{ cm}^3$, $l = 7 \text{ cm}$, $w = 8 \text{ cm}$.
Find h .

12 $V = 18 \text{ cm}^3$, $l = 3 \text{ cm}$, $w = 3 \text{ cm}$.
Find h .

YOU DO On your own: **Medium** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2 Questions 13–24: find the volume.

13 $6 \text{ m} \times 7 \text{ m} \times 4 \text{ m}$

14 $5.5 \text{ m} \times 5 \text{ m} \times 2 \text{ m}$

15 $1.4 \text{ m} \times 5 \text{ m} \times 5 \text{ m}$

16 $5.3 \text{ m} \times 5 \text{ m} \times 4 \text{ m}$

17 $4 \text{ m} \times 4 \text{ m} \times 2 \text{ m}$

18 $4.3 \text{ m} \times 6 \text{ m} \times 2 \text{ m}$

19 $5.8 \text{ m} \times 5 \text{ m} \times 4 \text{ m}$

20 $5.7 \text{ m} \times 5 \text{ m} \times 4 \text{ m}$

21 $5.4 \text{ m} \times 4 \text{ m} \times 3 \text{ m}$

22 $2.4 \text{ m} \times 8 \text{ m} \times 2 \text{ m}$

23 $4.9 \text{ m} \times 8 \text{ m} \times 4 \text{ m}$

24 $2.6 \text{ m} \times 5 \text{ m} \times 5 \text{ m}$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find the edge or the height.

a A cube has volume 64 cm^3 . Find its edge length.

Your teacher works this out. Copy every step.

b A box has volume 120 cm^3 , length 6 cm and width 4 cm. Find its height.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER Cube: $V = s^3$.

1 A shipping container is 12 m by 2.4 m by 2.6 m. Find its volume.

2 How many 5 cm cubes fit in a box 30 cm by 20 cm by 15 cm?

3 Sand is piled into a trailer 2 m by 1.2 m by 0.4 m. Find the volume of sand.



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

An L-shaped prism has a cross-section of area 90 m^2 and length 7 m . Find its volume.

1. $V = \text{area of cross-section} \times \text{length}$
2. 90×7

Answer: 630 m^3

YOU DO

On your own: read the worked example, then try these.

- E1** A prism has an L-shaped cross-section of area 45 cm^2 and length 12 cm . Find its volume.
- E2** A room is 5 m by 4 m by 2.7 m . An air conditioner cycles 1 m^3 of air every 2 seconds. How long to cycle all the air?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the volume: $5 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm}$.

Level 2

Find the volume of a cube with edge 6 m .

Level 3

A box has volume 72 cm^3 and base area 12 cm^2 . Find its height.

Lesson summary

Missed the lesson? Start here.

Remember 1. $V = \text{length} \times \text{width} \times \text{height}$.
3. Cube: $V = s^3$.

2. Or $V = \text{area of base} \times \text{height}$.
4. Answer in cubic units.

Level 1

$$8 \times 5 \times 2$$

1. 40×2

Answer: 80 m^3

Level 2

$$15 \times 10 \times 20$$

1. 150×20

Answer: 3000 cm^3

Level 3

$$\text{Cube } 64 \text{ cm}^3$$

1. $4 \times 4 \times 4 = 64$

Answer: 4 cm

Extension

L-prism: area 90, length 7

1. 90×7

Answer: 630 m^3

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
8.12

Volume of a prism

Date _____

Learning intentions

We are learning to:

- find the volume of prisms with triangular and other cross-sections

Success criteria

I can:

- ☐ find the area of the cross-section first
- ☐ use $V = A \times h$ (h is the length of the prism)
- ☐ use this for triangular, trapezium and composite prisms

Key terms Write the definition as your teacher explains it.

Cross-section

Prism

Notes

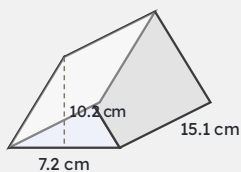
Level 1 · Getting started

Easy · 21 questions

WE DO With your teacher: copy what they write on the board.

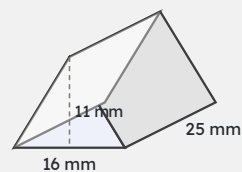
Example 1 Find the volume. $V = Ah$.

a Find the volume.



Your teacher works this out. Copy every step.

b Find the volume.



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **21 QUESTIONS**

REMEMBER Identify the cross-section (the end face).

ROUND 1 Questions 1–9: find the volume of the prism, $V = Ah$ (A = area of the cross-section, h = length).

1 $A = 58 \text{ cm}^2$, $h = 12 \text{ cm}$

2 $A = 5 \text{ cm}^2$, $h = 8 \text{ cm}$

3 $A = 11 \text{ cm}^2$, $h = 3 \text{ cm}$

4 $A = 44 \text{ cm}^2$, $h = 10 \text{ cm}$

5 $A = 22 \text{ cm}^2$, $h = 13 \text{ cm}$

6 $A = 7 \text{ cm}^2$, $h = 8 \text{ cm}$

7 $A = 28 \text{ cm}^2$, $h = 3 \text{ cm}$

8 $A = 57 \text{ cm}^2$, $h = 3 \text{ cm}$

9 $A = 17 \text{ cm}^2$, $h = 9 \text{ cm}$

YOU DO

On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2

Questions 10–21: triangular prism: find the area of the triangle, then the volume.

- 10 triangle $b = 12$ cm, $h = 3$ cm;
length 18 cm

- 11 triangle $b = 11$ cm, $h = 12$ cm;
length 7 cm

- 12 triangle $b = 2$ cm, $h = 3$ cm;
length 3 cm

- 13 triangle $b = 5$ cm, $h = 12$ cm;
length 14 cm

- 14 triangle $b = 6$ cm, $h = 10$ cm;
length 15 cm

- 15 triangle $b = 12$ cm, $h = 3$ cm;
length 12 cm

- 16 triangle $b = 10$ cm, $h = 7$ cm;
length 6 cm

- 17 triangle $b = 6$ cm, $h = 12$ cm;
length 5 cm

- 18 triangle $b = 12$ cm, $h = 7$ cm;
length 20 cm

- 19 triangle $b = 6$ cm, $h = 6$ cm;
length 7 cm

- 20 triangle $b = 12$ cm, $h = 2$ cm;
length 4 cm

- 21 triangle $b = 12$ cm, $h = 4$ cm;
length 7 cm

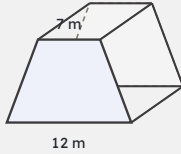
Level 2 · Building up

Medium · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find the volume. Find the area of the cross-section first.

- a** Find the volume (trapezium cross-section: parallel sides 7 m and 12 m, height 6 m, length 10 m).



Your teacher works this out. Copy every step.

- b** A prism has cross-section area 35 cm^2 and length 8 cm. Find its volume.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line (show working in the boxes).

18 QUESTIONS

REMEMBER Find its area A .

ROUND 1 Questions 1–12: the volume and the area of the cross-section are given. Find the length.

1 $V = 80 \text{ m}^3$, $A = 10 \text{ m}^2$

2 $V = 476 \text{ m}^3$, $A = 34 \text{ m}^2$

3 $V = 144 \text{ m}^3$, $A = 16 \text{ m}^2$

4 $V = 147 \text{ m}^3$, $A = 21 \text{ m}^2$

5 $V = 156 \text{ m}^3$, $A = 39 \text{ m}^2$

6 $V = 20 \text{ m}^3$, $A = 4 \text{ m}^2$

7 $V = 312 \text{ m}^3$, $A = 24 \text{ m}^2$

8 $V = 81 \text{ m}^3$, $A = 27 \text{ m}^2$

9 $V = 270 \text{ m}^3$, $A = 30 \text{ m}^2$

10 $V = 84 \text{ m}^3$, $A = 6 \text{ m}^2$

11 $V = 12 \text{ m}^3$, $A = 3 \text{ m}^2$

12 $V = 46 \text{ m}^3$, $A = 23 \text{ m}^2$

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 13–18: find the volume. Show your working.

- 13** triangular prism: base 10 m, height 9 m, length 8 m

- 14** triangular prism: base 8 m, height 2 m, length 17 m

- 15** triangular prism: base 8 m, height 3 m, length 22 m

- 16** triangular prism: base 11 m, height 5 m, length 5 m

- 17** triangular prism: base 3 m, height 4 m, length 21 m

- 18** triangular prism: base 4 m, height 2 m, length 14 m

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find the volume or the cross-section.

- a** Find the volume of a triangular prism: triangle base 6 cm, height 4 cm, prism length 10 cm.

Your teacher works this out. Copy every step.

- b** A prism has volume 360 m^3 and length 12 m. Find the area of its cross-section.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER $V = A \times \text{length of the prism}$.

- 1** A tent is a triangular prism: triangle base 2.4 m, height 1.5 m, length 3 m. Find its volume.

- 2** A ramp has a right-triangle cross-section with legs 1.2 m and 0.5 m and width 2 m. Find its volume.

- 3** A prism has volume 360 m^3 and length 12 m. Find the cross-section area.



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

A swimming pool is 25 m long and 10 m wide. It is 1 m deep at one end and 2 m deep at the other. Find its volume.

1. Cross-section (side view) is a trapezium: $\frac{1}{2} \times (1 + 2) \times 25 = 37.5 \text{ m}^2$
2. Width 10 m: 37.5×10

Answer: 375 m^3

YOU DO

On your own: read the worked example, then try these.

- E1 A pool 50 m by 20 m is 1.2 m deep at one end and 2 m at the other. Find its volume.

- E2 How many litres is the pool in the worked example? ($1 \text{ m}^3 = 1000 \text{ L}$)

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

A prism has cross-section 15 cm^2 and length 6 cm. Find its volume.

Level 2

Triangle base 5 m, height 4 m, prism length 9 m. Find the volume.

Level 3

A prism has volume 200 cm^3 and cross-section 25 cm^2 . Find its length.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Identify the cross-section (the end face).
 2. Find its area A .
 3. $V = A \times \text{length of the prism}$.
 4. Triangle: $A = \frac{1}{2}bh$. Trapezium: $A = \frac{1}{2}(a + b)h$.

Level 1

Triangle 7.2×10.2 ,
length 15.1

1. $A = \frac{1}{2} \times 7.2 \times 10.2 = 36.72$
2. $\times 15.1$

Answer: 554.472 cm^3

Level 2

$A = 35$, $h = 8$

1. 35×8

Answer: 280 cm^3

Level 3

Trapezium 7, 12, height
6, length 10

1. $\frac{1}{2} \times 19 \times 6 = 57$
2. $\times 10$

Answer: 570 m^3

Extension

Sloping pool

1. 37.5×10

Answer: 375 m^3

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
8.13

Capacity

Date _____

Learning intentions

We are learning to:

- find the capacity of a container in litres

Success criteria

I can:

- ☐ know $1 \text{ cm}^3 = 1 \text{ mL}$ and $1 \text{ m}^3 = 1000 \text{ L}$
- ☐ find volume, then convert to litres
- ☐ solve problems about water levels and large volumes

Key terms Write the definition as your teacher explains it.

Capacity

Megalitre

Notes

Level 1 · Getting started

Easy · 28 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Find the capacity, then the water height.

- a** A box is 30 cm by 20 cm by 10 cm. Find its capacity in litres.

Your teacher works this out. Copy every step.

- b** The water is poured into a prism with base 20 cm by 20 cm. How high does it reach?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

28 QUESTIONS

REMEMBER $1 \text{ cm}^3 = 1 \text{ mL}$; $1000 \text{ cm}^3 = 1 \text{ L}$.**ROUND 1** Questions 1–12: convert. 1 L = 1000 mL, 1 kL = 1000 L.

1 50 L to kL

2 1 600 mL to L

3 5 kL to L

4 19 L to mL

5 58 kL to L

6 0.3 L to mL

7 57 kL to L

8 0.58 L to mL

9 0.42 L to mL

10 3.9 kL to L

11 17 000 mL to L

12 18 000 mL to L

YOU DO On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2 Questions 13–28: convert. $1 \text{ cm}^3 = 1 \text{ mL}$, $1000 \text{ cm}^3 = 1 \text{ L}$.

- | | | | |
|-------------------------------------|------------------------------------|-------------------------------------|-------------------------------------|
| 13 4.3 L to cm^3
_____ | 14 0.8 L to cm^3
_____ | 15 3900 cm^3 to L
_____ | 16 1.2 L to cm^3
_____ |
| 17 3500 cm^3 to L
_____ | 18 3.8 L to cm^3
_____ | 19 4.8 L to cm^3
_____ | 20 1 L to cm^3
_____ |
| 21 1.7 L to cm^3
_____ | 22 2.8 L to cm^3
_____ | 23 3200 cm^3 to L
_____ | 24 4100 cm^3 to L
_____ |
| 25 1800 cm^3 to L
_____ | 26 4.4 L to cm^3
_____ | 27 3100 cm^3 to L
_____ | 28 2.3 L to cm^3
_____ |

Level 2 · Building up

Medium · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Convert, then find the capacity.

a Convert 3.5 m^3 to litres.

Your teacher works this out. Copy every step.

b Find the capacity in litres of a tank 8 m by 5 m by 2 m.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

18 QUESTIONS

REMEMBER $1 \text{ m}^3 = 1000 \text{ L} = 1 \text{ kL}$.

ROUND 1 Questions 1–12: find the capacity of the box in litres.

- | | | |
|---|---|---|
| 1 $50 \text{ cm} \times 40 \text{ cm} \times 5 \text{ cm}$
_____ | 2 $50 \text{ cm} \times 50 \text{ cm} \times 5 \text{ cm}$
_____ | 3 $50 \text{ cm} \times 50 \text{ cm} \times 25 \text{ cm}$
_____ |
| 4 $20 \text{ cm} \times 30 \text{ cm} \times 20 \text{ cm}$
_____ | 5 $60 \text{ cm} \times 40 \text{ cm} \times 25 \text{ cm}$
_____ | 6 $40 \text{ cm} \times 20 \text{ cm} \times 20 \text{ cm}$
_____ |
| 7 $60 \text{ cm} \times 10 \text{ cm} \times 15 \text{ cm}$
_____ | 8 $40 \text{ cm} \times 40 \text{ cm} \times 15 \text{ cm}$
_____ | 9 $30 \text{ cm} \times 20 \text{ cm} \times 20 \text{ cm}$
_____ |
| 10 $30 \text{ cm} \times 10 \text{ cm} \times 10 \text{ cm}$
_____ | 11 $20 \text{ cm} \times 50 \text{ cm} \times 25 \text{ cm}$
_____ | 12 $50 \text{ cm} \times 40 \text{ cm} \times 10 \text{ cm}$
_____ |

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 13–18: how many can be filled?

13 500 mL cups from 9 L

14 250 mL cups from 6 L

15 250 mL cups from 9 L

16 200 mL cups from 9 L

17 200 mL cups from 2 L

18 300 mL cups from 9 L

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve.

a An iceberg is like a prism 160 km by 50 km by 250 m. How many megalitres of water?

Your teacher works this out. Copy every step.

b A 2 L jug fills cups of 250 mL. How many cups?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER 1 ML (megalitre) = 1 000 000 L.

1 A 30 cm × 20 cm × 10 cm box of water is poured into a prism with base 20 cm × 20 cm. How high does the water reach?

2 Iceberg B9 was like a prism 160 km by 50 km by 250 m. How many megalitres of water did it hold?

3 A fish tank 60 cm × 30 cm × 40 cm is filled to 5 cm from the top. How many litres of water?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Convert 2500 cm³ to litres.

Level 2

Find the capacity in litres of a box 25 cm × 20 cm × 10 cm.

Level 3

Convert 0.6 m³ to litres.

Extension · Stretch yourself**Worked example**

Rain of 10 mm falls on a 12 m by 8 m roof. How many litres run into the tank?

1. $10 \text{ mm} = 0.01 \text{ m}$
2. $V = 12 \times 8 \times 0.01 = 0.96 \text{ m}^3$
3. $\times 1000$

Answer: 960 L

YOU DO

On your own: read the worked example, then try these.

- E1** How much rain (mm) must fall on the roof to fill a 4800 L tank?

- E2** A house uses 180 L per person per day. How many days does 960 L last a family of 4?

Lesson summary

Missed the lesson? Start here.

Capacity 1. $1 \text{ cm}^3 = 1 \text{ mL}$; $1000 \text{ cm}^3 = 1 \text{ L}$.
3. 1 ML (megalitre) = 1 000 000 L.

2. $1 \text{ m}^3 = 1000 \text{ L} = 1 \text{ kL}$.
4. Find the volume first, then convert.

Level 1

$30 \times 20 \times 10 \text{ cm}$

1. 6000 cm^3

Answer: 6 L

Level 2

Pour into 20×20 base

1. $6000 \div 400$

Answer: 15 cm (D)

Level 3

$8 \times 5 \times 2 \text{ m}$

1. $80 \text{ m}^3 \times 1000$

Answer: 80 000 L

Extension

Rain on a roof

1. 0.96 m^3

Answer: 960 L

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